

Mineralogic and Stable Isotope Studies of Hydrothermal Alteration at the Jinchuan Ni-Cu Deposit, China

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Abstract

The Jinchuan intrusion hosts one of the largest magmatic Ni-Cu sulfide deposits in the world. Net-textured and disseminated pyrrhotite, pentlandite, and chalcopyrite are found in olivine-rich rocks (60–80 vol %) that occur as a northwest-southeast-striking dike. Olivine in the intrusion has been variably converted to mixtures of serpentine and fine-grained magnetite. Interstitial pyroxene and minor plagioclase have been altered to mixtures of chlorite, amphibole, epidote, clinozoisite, Na-rich feldspar, and lesser amounts of calcite. Although the **interstitial** net texture of sulfide minerals surrounding olivine grains is preserved, up to 30 vol percent of sulfide minerals are replaced by magnetite, serpentine, and chlorite. An external origin for Mg and Si needed for the isovolumetric replacement of sulfides cannot be discounted, but a local origin related to olivine and pyroxene alteration is favored. Serpentine and chlorite may be enriched in Cu and Ni, but the enrichment can account for less than 1 percent of the metals lost during the replacement of sulfide minerals by silicates and oxides. Some of the metals and sulfur that were removed from the net-textured sulfide assemblages have been redeposited on a centimeter scale in fine ribbons and veins within serpentine, chlorite, and amphibole.

Sulfur isotope values of the pyrrhotite-pentlandite-chalcopyrite assemblage at Jinchuan fall in the range of –2 to 8 per mil, with over 80 percent of the values between –2 and 2 per mil. Immediate country rocks in the Jinchuan area are Precambrian granitoids, schists, and marbles with sulfur concentrations less than 100 ppm. Although the $\delta^{34}\text{S}$ values of the main sulfide assemblage at Jinchuan are in the range that is characteristic of mantle-derived sulfur, assimilation of sulfur from unexposed country rocks cannot yet be discounted. Secondary pyrite occurs in veins and stringers, and rarely as a replacement of pyrrhotite, and is characterized by $\delta^{34}\text{S}$ values between –7 and –27 per mil. The strongly negative $\delta^{34}\text{S}$ values of secondary pyrite are thought to be a result of oxidation of primary magmatic sulfide assemblages and partial reduction of aqueous sulfate.

Oxygen and hydrogen isotope values of silicate minerals suggest that at least two fluids have been involved in the hydrothermal alteration at Jinchuan. The $\delta^{18}\text{O}$ values of plagioclase, amphibole, and serpentine all suggest the involvement of a fluid with a $\delta^{18}\text{O}$ value between –2 and 5 per mil at 300° to 400°C. The δD values of amphibole and one population of serpentine suggest a δD value for the fluid in the range of –45 to –63 per mil. These values are consistent with the involvement of evolved seawater or a metamorphic fluid in the early stages of alteration. The possible involvement of evolved seawater in hydrothermal alteration is consistent with a rift setting for the emplacement of the Jinchuan intrusion. However, the wide range in serpentine δD values (computed δD of the fluid = –45 to –103‰) suggest either that serpentinization involved a mixture of evolved seawater and/or metamorphic water and meteoric water or, more likely, that serpentine continued to exchange hydrogen isotopes with late-stage meteoric waters after it formed.

Introduction

THE JINCHUAN Ni-Cu deposit in northwestern China is one of the world's largest magmatic deposits, with reserves, of 500 million metric tons (Mt) of sulfide ore, with grades of 1.06 percent Ni and 0.7 percent Cu (Chai and Naldrett, 1992a). Previous studies of the deposit (e.g., Tang, 1990, 1993; Chai and Naldrett, 1992b; Tang and Li, 1995; Li, C. et al., 2004; De Waal et al., 2004) have focused on the primary igneous petrogenesis of the Jinchuan intrusion and the development of models for sulfide ore genesis. Hydrothermal alteration and/or metamorphism that has affected the intrusion and sulfide ores was recognized by Chai and Naldrett (1992b), Barnes and Tang (1999), De Waal et al. (2004), and Li, C. et al. (2004), but no detailed studies that address these secondary processes have been undertaken. Here we report the results of mineralogical and stable isotope studies of the deposit that bear on secondary processes that have affected the intrusion and associated mineralization.

Geologic Background

The Jinchuan deposit occurs in a dikelike body (~6,500 m long, <500 m wide, and >1,100 m deep in its central part) dominated by olivine-rich rocks such as harzburgite, lherzolite, and dunite, with lesser amounts of norite, gabbro, and gabbro-norite. The sulfide mineralization is hosted by the Jinchuan intrusion. The sulfides mainly occur in the form of disseminated and net-textured pyrrhotite, chalcopyrite, and pentlandite, which commonly comprise between 2 and 30 vol percent of the olivine-rich rocks. Massive sulfide mineralization is less common. The deposit is divided into three segments by a series of northeast-trending strike-slip faults (Fig. 1).

The Jinchuan intrusion occurs along the southwestern margin of the Sino-Korean platform (Fig. 1). Host rocks to the Jinchuan intrusion are part of the Proterozoic Baijiazui Formation (Fig. 1) and include granulite facies marble, gneiss, and granitic migmatite. Both the age and metamorphic grade of the Proterozoic sequence decrease to the southwest. Rocks of the uppermost Proterozoic unit include sericite schist, carbonate schist, limestone, and intercalated andesitic metavolcanics (Chai and Naldrett, 1992b). The northern boundary of

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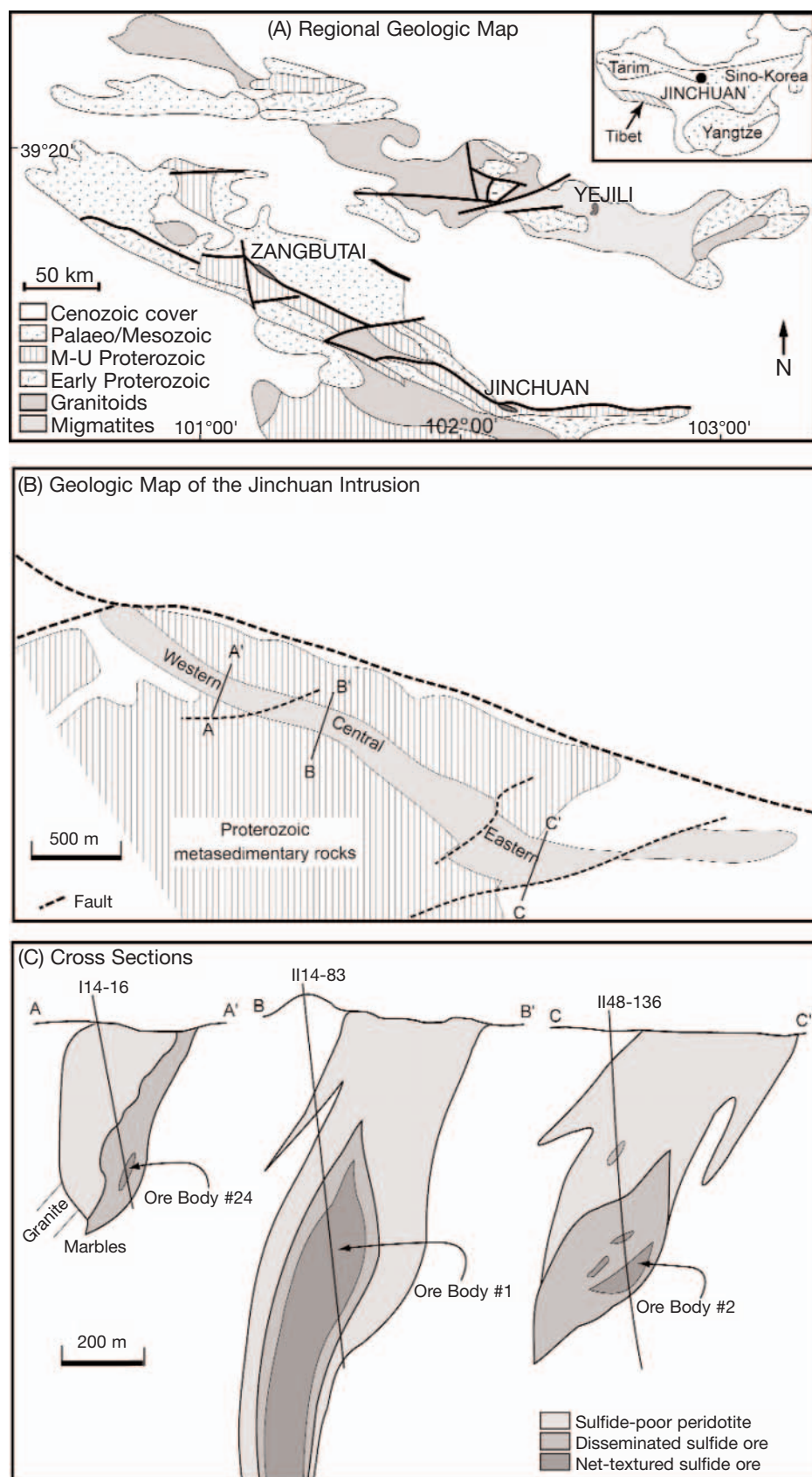


FIG. 1. A. Location and geologic map of the Jinchuan area. B. Cross sections based on at least five drill cores. Only the sampled cores are illustrated (after Li et al., 2004).

the intrusive sequences is a fault contact with post-Paleozoic sedimentary rocks (Fig. 1). Recent U-Pb SHRIMP dating of zircon and baddeleyite from the Jinchuan intrusion by Li, X. et al. (2004, 2005) indicate a crystallization age of 827 ± 8 Ma and a possible relationship with mantle plume activity that coincided with the breakup of Rodinia.

Previous studies of the deposit have led to conflicting theories with respect to igneous evolution and sulfide genesis. Chai and Naldrett (1992b) proposed that the Jinchuan intrusion represented the root zone of a much larger layered intrusion and that the dike-like central part of the intrusion was a feeder to the magma chamber. Tang (1993) and Tang and Li (1995) suggested that the intrusion formed by multiple inputs from a deep-seated stratified magma chamber. Sulfide-bearing magma was proposed to have been extracted from the lower portion of the deep chamber, whereas sulfide-barren intrusions were extracted from the upper portion. De Waal et al. (2004) and Li, C. et al. (2004) showed via trace element and olivine compositional data that emplacement involved high-viscosity, dense crystal-rich mushes. Several pulses of magma are required to explain the variations in olivine composition, and flow and gravitational differentiation of a sulfide-rich magma are thought to account for the present distribution of textural features. Like Tang (1993) and Tang and Li (1995), De Waal et al. (2004) call upon emplacement of sulfide-bearing and sulfide-free pulses from a deep, stratified magma chamber to explain the occurrence of sulfide-barren as well as sulfide-bearing rocks in the Jinchuan area.

Sampling and Analytical Methods

Samples were collected from three drill cores from the western, central, and eastern portions of the intrusion (Fig. 1). A total of 120 samples have been examined petrographically. Olivine, serpentine, amphibole, and plagioclase were isolated for oxygen and hydrogen isotope analyses using a 0.75-mm diamond-tipped microdrill. Sulfide minerals for sulfur isotope measurements were sampled using a carbide-tipped microdrill. Olivine concentrates contained up to 20 percent serpentine and serpentine separates contained as much as 10 percent finely dispersed magnetite, determined by polarized light microscopy.

Sulfide mineral separates were analyzed using elemental analyzer-continuous flow isotope ratio mass spectrometry (Studley et al., 2002). Sample powders (~0.1 mg) were placed in tin cups with 1 to 1.5 mg of V_2O_5 and combusted at ~1,800°C, with a reactor column temperature of 1,010°C. Analyses of SO_2 were made using a Finnigan MAT 252 stable isotope ratio mass spectrometer, with results reported in standard delta notation relative to VCDT. Instrumental uncertainty for each analysis was less than ± 0.05 per mil, and sample reproducibility (1–3 replicates of samples from the same thin section blank) was within ± 0.3 per mil. Sulfide standards utilized included IAEA S-1 (with a defined value of -0.3‰ VCDT), IAEA S-2 (21.7‰), IAEA S-3 (-31.3‰) and several Indiana University laboratory standards.

Oxygen from silicate minerals was extracted using the BrF_5 method of Clayton and Mayeda (1963). Oxygen was converted to CO_2 by reaction with a heated graphite disk. Isotopic analyses were made using a Finnigan MAT 252 stable isotope ratio mass spectrometer, with results reported in stan-

dard delta notation relative to VSMOW. Instrumental uncertainty was less than 0.05 per mil, and sample reproducibility was ± 0.1 to ± 0.2 per mil. The value of NBS-28 quartz in our lab is 9.6 ± 0.2 per mil.

Samples for hydrogen isotope analyses were analyzed using a high-temperature-elemental analyzer-continuous flow method similar to that reported by Sharp et al. (2001). Samples were loaded into silver cups and combusted at 1,400°C in a glassy carbon column. Hydrogen was analyzed using a Finnigan Delta Plus-XP stable isotope ratio mass spectrometer. Several standards were run with each series of analyses. Internal standards ranged from -60 to -100 per mil and were calibrated relative to an NBS-30 biotite value of -65 per mil. Results are reported in delta notation with respect to VSMOW. Instrumental reproducibility was within ± 0.2 per mil and sample reproducibility was ± 2 per mil.

Estimates of sulfide and silicate mineral proportions in grains where replacement was observed were made using polished thin sections and digital images. Digital photomicrographs were converted to gray scale and areas corresponding to specific minerals were measured using GIS area-volume software.

Backscattered electron imaging and quantitative analyses of minerals were conducted using the CAMECA SX50 electron microprobe at Indiana University. Mineral chemical analyses were determined by wavelength-dispersive X-ray analysis using an acceleration voltage of 15 kV, a beam current of 15 nA, and a counting time of 20 s. Copper and Ni were analyzed at a beam current of 50 nA and a counting time of 120 s. Detection limits for Cu and Ni in serpentine and chlorite were ~150 ppm.

Results

Petrography and compositional variations

Most of the samples in this study contain olivine in excess of 60 vol percent and would be classified as dunite to lherzolite. Interstitial minerals include orthopyroxene, plagioclase, clinopyroxene, sulfides, and phlogopite. Only phlogopite appears not to have been affected by hydrothermal alteration; all other minerals show varying degrees of conversion to secondary minerals. Typical alteration products are serpentine and fine-grained magnetite, derived from olivine, chlorite, and amphibole, derived from pyroxene, and chlorite, epidote, zoisite, and more sodic feldspar derived from plagioclase. X-ray diffraction analyses indicate that the most abundant serpentine mineral present is antigorite, but lizardite and chrysotile may be present as well in small amounts. More detailed studies involving transmission electron microscopy are necessary to refine the serpentine mineralogy. Although hydrothermal alteration is spatially ubiquitous, not all samples show an equal intensity of alteration. The most extensive hydrothermal alteration occurs along sheared contacts with country rocks, but there is not a uniform decrease in the degree of alteration toward the center of the intrusion (Fig. 2). In some samples olivine is only partially serpentinized, and large grains with only small serpentine-filled fractures are present. By far the most common texture is a mesh of serpentine and magnetite enclosing remnant olivine kernels (Fig. 3A). In a few samples from the western portion

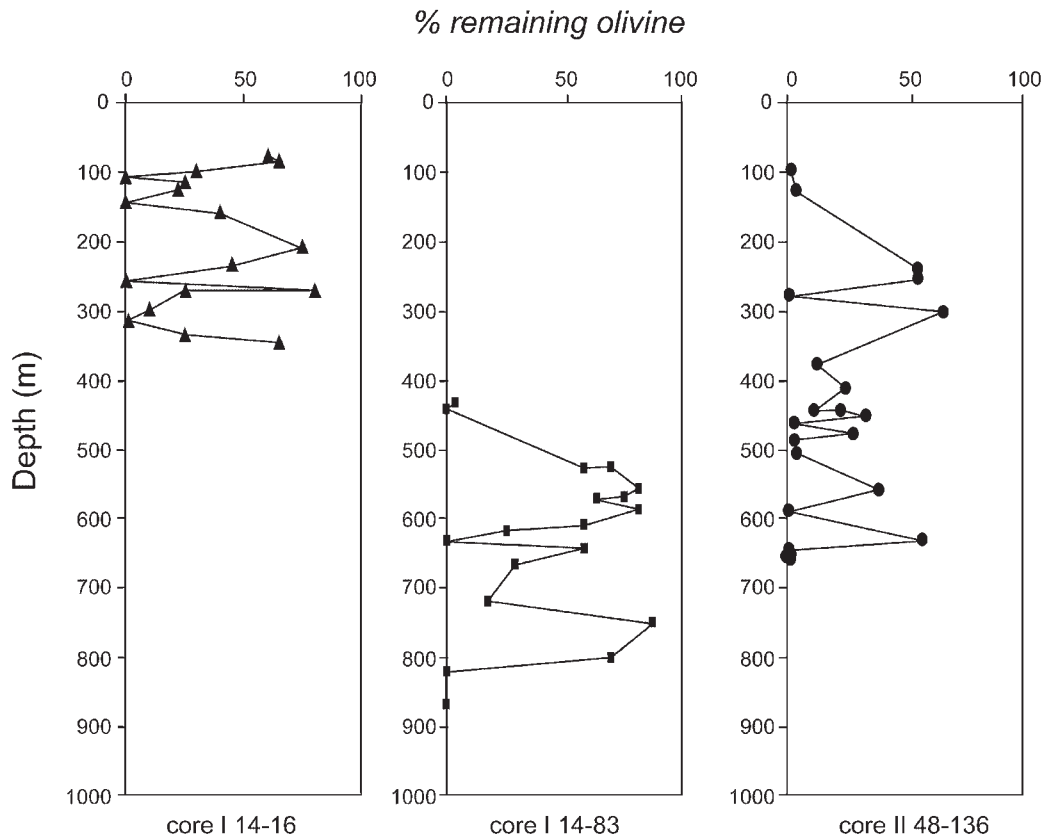


FIG. 2. Plot showing the percentage of olivine remaining in thin sections from sampled drill cores where pseudomorphic replacement of olivine by serpentine, chlorite, magnetite, and amphibole has occurred.

of the deposit (drill core I 14-16; Fig. 1), unaltered orthopyroxene is found enclosing grains of generally fresh to partially serpentinized, subrounded olivine. Serpentine occurs in some of the included olivine grains, but the surrounding orthopyroxene is unaffected (Fig. 3B). Although interstitial plagioclase is normally strongly altered to mixtures of chlorite, epidote, calcite, and albite, areas of turbid but unreplaced plagioclase occur locally. Li, C. et al. (2004)

report that, within sheared intervals along contacts with country rocks, rims of olivine in contact with chlorite and Mg hornblende are recrystallized to more Fe-rich compositions. In more intensely altered samples from the shear zones, tremolite replaces Mg hornblende, and the size of the recrystallized olivine rim increases. In the most advanced stages of alteration in the shear zones a mesostasis of actinolite needles is present with completely recrystallized Fe-rich

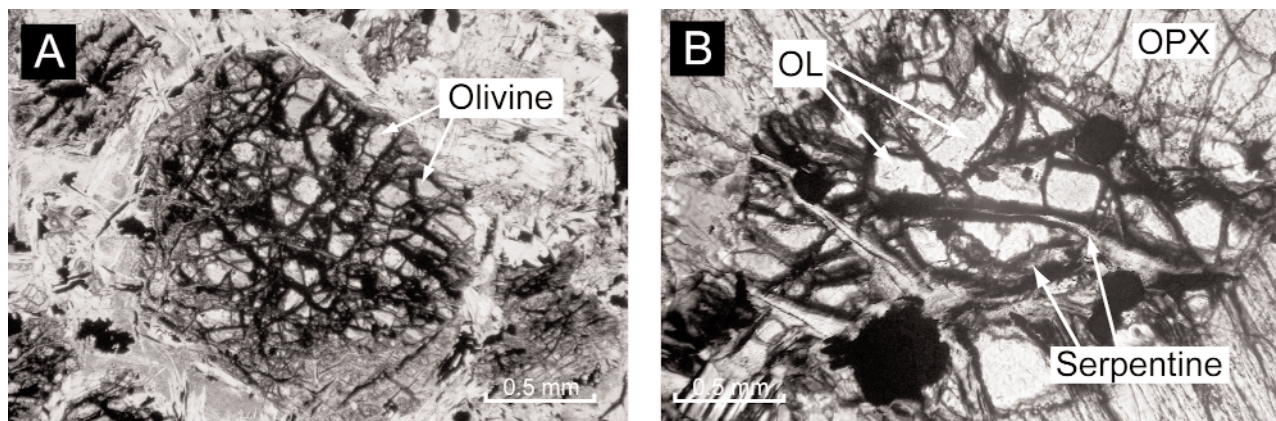


FIG. 3. A. Remnant olivine replaced by serpentine and magnetite, in a matrix of chlorite plus amphibole (transmitted, plane-polarized light, sample II14-32). B. Serpentinized olivine grain in orthopyroxene. Note that the serpentine in fractures within the olivine does not penetrate the enclosing orthopyroxene (transmitted light, crossed polars, sample II 14-12).

olivine. Minor amounts of talc and magnetite are also present in the most intensely altered samples from the shear zones.

Virtually all of the samples examined contain sulfide minerals. Interstitial sulfides are commonly finely disseminated together with other interstitial minerals between olivine grains; the abundance of sulfides is gradational up to ~30 vol percent. Where sulfides are most abundant, a characteristic net texture is developed with almost no other interstitial minerals (Fig. 4A-B). The average size of the sulfide-filled interstices is between 1 and 2 mm. Sulfide minerals include pyrrhotite, chalcopyrite, and pentlandite generally in the proportion 70:25:5. The same sulfide minerals in similar proportions also may occur as very fine ribbons or veinlets within serpentine, chlorite, and amphibole. These thin fracture-filling sulfides may or may not be physically connected to interstitial, net-textured sulfide minerals. Triple-junction boundaries between subgrains, characteristic of recrystallization and annealing (e.g., Stanton, 1972), are locally found in some sulfide grains. This texture is most commonly found within thin vein-type chalcopyrite or pyrrhotite that link net-textured areas.

Although in hand samples the net-textured sulfides appear to form solid masses surrounding olivine grains, when examined microscopically all samples show variable development

of veins in sulfides which are composed of magnetite, serpentine, chlorite, calcite, and locally amphibole. The interstitial net-textured form of the sulfide is generally preserved with no deformation of the enclosed olivine grains (Fig. 4C-D), suggesting isovolumetric replacement of sulfides by secondary magnetite and silicates. Where the intensity of alteration is low, silicate minerals and magnetite in the sulfides occur in relatively widely spaced veins (Fig. 5A-B), and with increased alteration intensity a more mesh-like network with fine-grained remnant sulfide between the silicate minerals and magnetite is developed. The vein and network pattern within pentlandite results in a much more dense pattern of replacement of sulfides by silicates than in either chalcopyrite or pyrrhotite (Fig 6A). Magnetite may occur as euhedral cubic to octahedral grains that are thought to be part of the primary magmatic assemblage, as fine ribbons to veins in recrystallized and replaced sulfide minerals (Fig. 6B), or as rims on partially replaced sulfides. The latter two types of magnetite are clearly of secondary origin. Goethite also may occur as rims on secondary magnetite or sulfides. Pyrite has been identified in several samples from Jinchuan, but it is generally not part of the pyrrhotite-chalcopyrite-pentlandite assemblage. Pyrite cubes occur within pyrite-carbonate $\pm$ magnetite $\pm$ serpentine and chlorite veins (typically 1–2 mm in width) that crosscut silicate and sulfide assemblages (Fig.

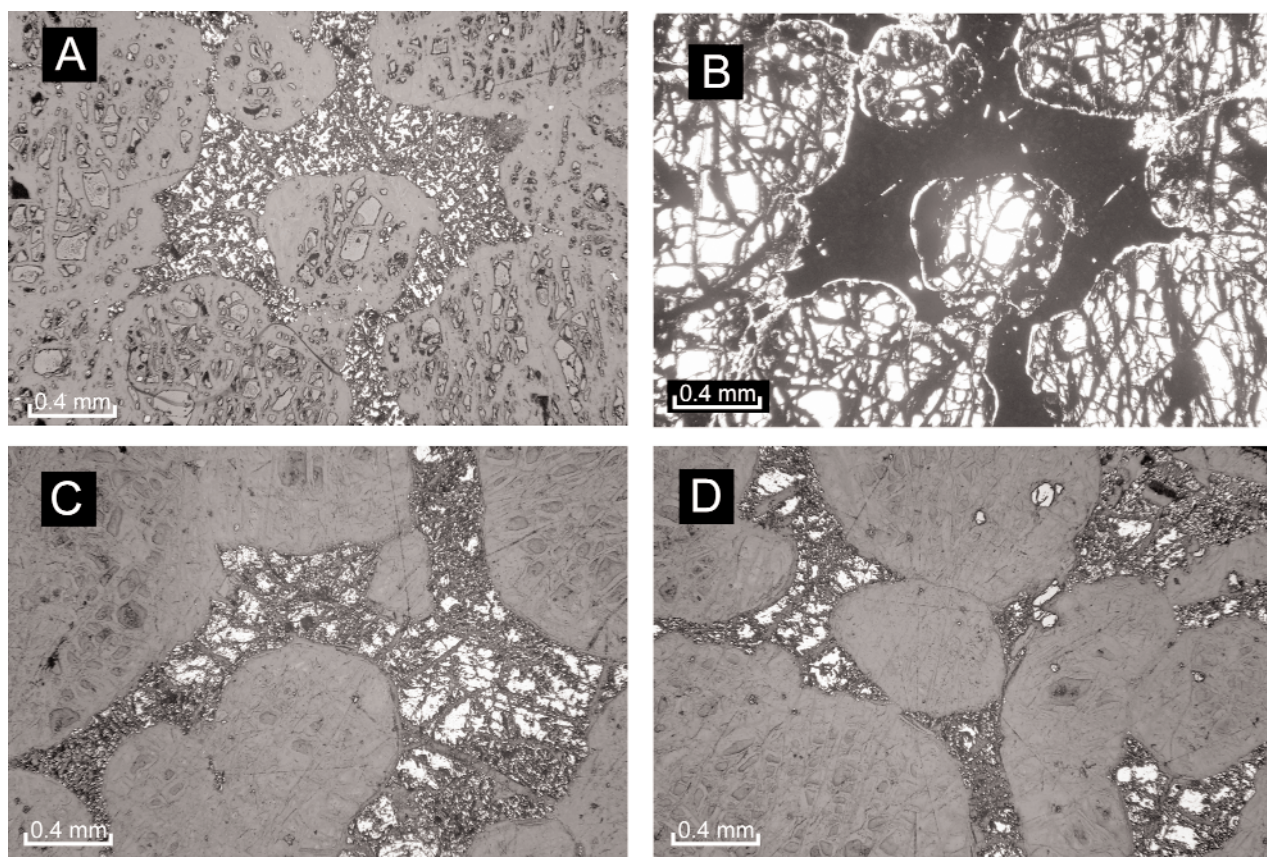


FIG. 4. A. Serpentinized olivine enclosed by net-textured sulfide minerals (reflected, plane-polarized light, sample II-14-30). B. Same field as (A) but transmitted plane-polarized light. C. and D. Net-textured sulfides with fracture-related serpentine, chlorite, and magnetite (dark areas) around serpentinized olivine (reflected, plane-polarized light, sample II 14-21).

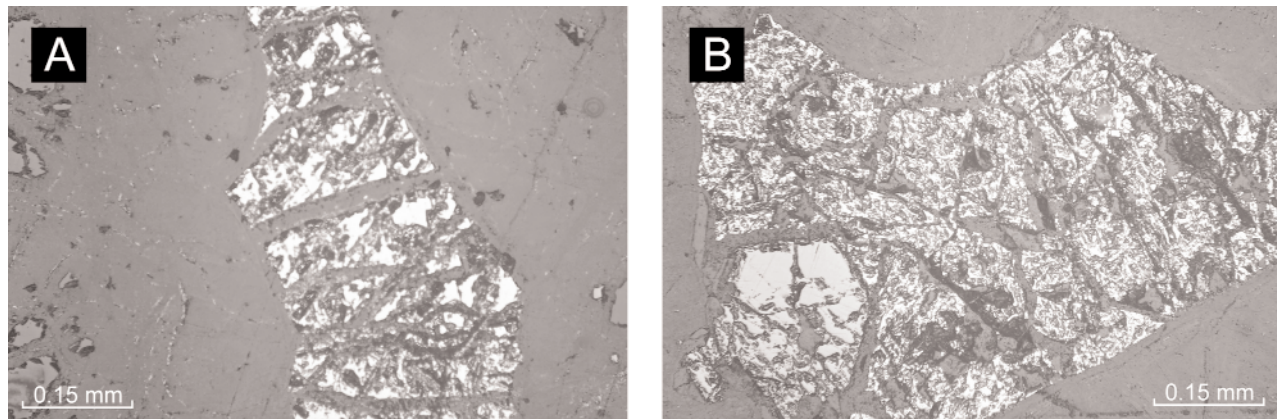


FIG. 5. A. Interstitial, net-textured pyrrhotite extensively replaced by veinlike serpentine and chlorite (reflected, plane-polarized light, sample II 14-30). B. Interstitial net-textured pyrrhotite, chalcopyrite, and magnetite (mt), veined and replaced by serpentine plus chlorite (reflected, plane-polarized light, sample II 14-21).

6B). Where the percentage of pyrite-bearing veins is high, pyrite replaces net-textured pyrrhotite-rich assemblages (Fig. 6B).

Compositional variations of serpentine and chlorite that replace various sulfide minerals and olivine in two samples are given in Table 1. Serpentine that forms a mesh texture around olivine contains between 1,000 and 2,000 ppm Ni, whereas that which occurs in pentlandite contains ~3,000 ppm Ni. Serpentine within pyrrhotite contains less than 650 ppm Ni and that within chalcopyrite has less than 150 ppm. A sample

of chlorite within fractures in pentlandite also contains an average of ~1,800 ppm Ni. In one sample, the Cu content of serpentine that replaces chalcopyrite averages ~750 ppm. Careful electron microprobe analyses were done using a focused beam to avoid potential contribution from surrounding chalcopyrite. Although no inclusions or sulfide stringers were detected, it is possible that Cu was contributed from submicroscopic sulfide inclusions. Ion-microprobe and TEM analyses may be required to better evaluate the unusually high Cu content of this serpentine.

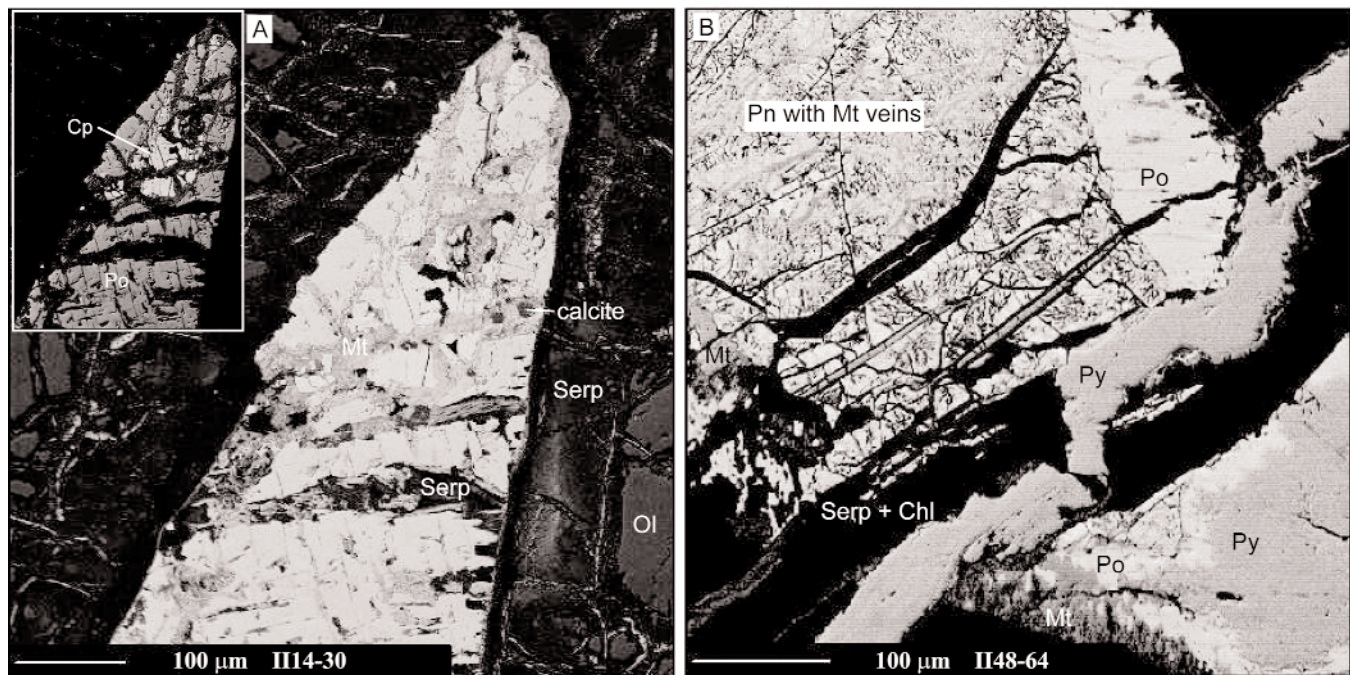


FIG. 6. Backscattered electron images of samples from the Jinchuan deposit. A. A pyrrhotite-chalcopyrite grain showing sharp contacts with adjoining serpentinized olivine. The grain is cut by fracture-related magnetite and serpentine. Although the sharp outline of the grain is preserved, secondary minerals have developed along fracture zones. B. A pyrrhotite-pentlandite composite grain showing the high density of fracture-filling magnetite and serpentine in pentlandite. The fine dark-colored dendritic areas are serpentine within microfractures. Pyrite and magnetite replace pyrrhotite in the lower right. The grain is cut by a late-stage vein of pyrite + serpentine + chlorite.

TABLE 1. Representative Electron Microprobe Analyses of Serpentine and Chlorite from the Jinchuan Deposit

Sample	Type		Wt percent								Parts per million	
Serpentine		<i>n</i>	MgO	Al ₂ O ₃	SiO ₂	MnO	FeO	CaO	H ₂ O ¹	Total	Cu	Ni
II 14-83/722m	Replacing olivine	5	38.96	0.34	43.08	0.13	5.12	0.06	12.81	100.49	nd	1014
II 14-83/610m	Replacing olivine	4	41.21	0.03	39.04	0.09	5.99	0.03	12.41	98.79	nd	2042
II 14-83/610m	Replacing pentlandite	5	40.42	0.03	40.17	0.13	5.79	0.07	12.51	99.11	nd	2906
II 14-83/610m	Replacing pyrrhotite	6	40.00	0.06	39.68	0.13	7.30	0.05	12.47	99.69	nd	639
II 14-83/722m	Replacing chalcopyrite	3	41.87	0.49	41.41	0.02	2.98	0.07	12.51	99.34	752	nd
Chlorite												
II 14-83/610m	Replacing pentlandite	3	33.26	16.57	31.84	4.43	0.06	1.60	12.72	100.76	nd	1795

Abbreviation: nd = not detected

¹ By stoichiometry

Stable isotopes

The $\delta^{34}\text{S}$ measurements of sulfide minerals are presented in Table 2 and Figure 7. The majority (>80%) of all samples have $\delta^{34}\text{S}$ values in the range of -2 to +2 per mil. All of the samples with this isotopic composition are from interstitial and net-textured pyrrhotite-chalcopyrite-pentlandite assemblages. Our initial analyses indicated that no isotopic differences existed between pyrrhotite and chalcopyrite, and for this reason no attempt was made to separate these minerals. A few samples from the pyrrhotite-chalcopyrite-pentlandite assemblage have elevated $\delta^{34}\text{S}$ values between 3 and 8 per mil. Pyrite from the pyrite-carbonate veins has negative $\delta^{34}\text{S}$ values between -7 and -27 per mil. Samples of net-textured sulfides where pyrite has replaced pyrrhotite have $\delta^{34}\text{S}$ values in the range of -3 to -5 per mil.

The $\delta^{18}\text{O}$ and δD values of olivine, plagioclase, serpentine, amphibole, and whole rocks (δD) are presented in Table 2. The $\delta^{18}\text{O}$ values are as follows: olivine (3.4–5.9‰), plagioclase (1.1–3.1‰), amphibole (2.4–4.1‰), and serpentine (2.1–3.6‰). The δD values range from -75 to -123 per mil

(serpentine), -75 to -86 per mil (amphibole), and from -73 to -125 per mil (whole rocks).

Discussion

Temperatures of hydrothermal alteration

Temperatures of mineral reactions in the Jinchuan area are difficult to estimate. Although we have searched for fluid inclusions in calcite veins, the high density of fracturing apparently leads either to a lack of preservation or the presence of extremely small inclusions that are unworkable. The absence of prehnite and pumpellyite in the plagioclase-bearing samples suggests that temperatures were above those of the prehnite-pumpellyite to greenschist facies transition, determined by Cho and Liou (1987), Frey et al. (1991), and Digel and Gordon (1995) to be less than ~275°C at pressures less than ~2 kbars. Experimental studies indicate that at pressures of ≤500 bars and temperatures below ~330°C, Fe-bearing olivine reacts to produce serpentine, brucite, and magnetite (Moody, 1976; Janecky and Seyfried, 1986; Allen and Seyfried, 2003). Interstitial clinopyroxene has been altered to mixtures of chlorite and actinolite, which Wicks (1984) and O'Hanley (1996) reported can occur when hydrothermal alteration occurs at temperatures below 400°C and pressures of ~500 bars. Because of disequilibrium during isotopic exchange, oxygen isotope data from mineral pairs are not a reliable indication of temperature. Based on the absence of prehnite and pumpellyite, plus the presence of partially serpentinized olivine and highly altered pyroxene, a temperature range between 275° and 400°C appears to be reasonable for the pervasive alteration in the Jinchuan area.

Mechanisms of sulfide mineral replacement

Mineralogical alteration at Jinchuan is mainly controlled by microfractures as evidenced by the development of serpentine-magnetite mesh texture in olivine and the occurrence of magnetite, serpentine, and chlorite veins in sulfide minerals. The replacement of sulfide minerals by serpentine has been reported previously by Frost (1985), O'Hanley and Wicks (1995), and Barnes and Hill (2000). We interpret the serpentine ± chlorite and magnetite veins in sulfide minerals at Jinchuan to similarly represent a near-isovolumetric replacement process driven by local fluid infiltration and solution and/or redeposition along microfractures. Isovolumetric replacement is a process whereby the rate of dissolution is

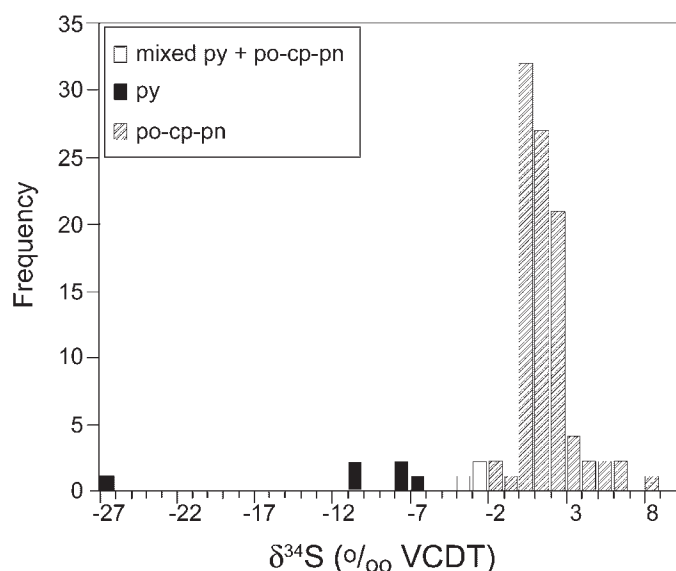


FIG. 7. Histogram of $\delta^{34}\text{S}$ values from the Jinchuan deposit showing the different ranges for pyrite and pyrrhotite populations.

TABLE 2. Stable Isotope Data From the Jinchuan Deposit

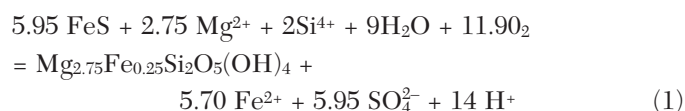
Drill core	Depth (m)	VCDT (‰)		VSMOW (‰)						
		$\delta^{34}\text{S}$ (po-cp)	$\delta^{34}\text{S}$ (py)	$\delta^{18}\text{O}$ (ol)	$\delta^{18}\text{O}$ (pl)	$\delta^{18}\text{O}$ (se)	$\delta^{18}\text{O}$ (am)	δD (se)	δD (am)	δD (wr)
I 14-16	60	0.8								
I 14-16	77	2.2								-100
I 14-16	85	6.0								-94
I 14-16	106	0.8								
I 14-16	114		-8.8							
I 14-16	124		-11.3							
I 14-16	133	-0.3				2.6		-90		
I 14-16	143	0.5								
I 14-16	153	-0.7								
I 14-16	158	-2.6								-125
I 14-16	175				3.1			-81		
I 14-16	195	0.3						-70		
I 14-16	209	6.0		5.0	2.2					
I 14-16	220	1.6								
I 14-16	234	2.5		4.9			2.7	-92	-86	
I 14-16	257	5.4								-109
I 14-16	271	-2.3								
I 14-16	291	0.1				3.0	2.4	-105	-79	
I 14-16	299	1.2								
I 14-16	306		-27.0							
I 14-16	315	0.8								
I 14-16	330	0.3								
I 14-16	335	0.6								
I 14-16	335	2.1								
I 14-16	345	0.8								
I 14-16	366	4.9								
I 14-16	377	0.2								
II 14-83	360	-0.1								
II 14-83	370	4.8								
II 14-83	435	1.9		5.9			4.1	-75	-75	
II 14-83	459	2.8								
II 14-83	473	2.7								
II 14-83	483	0.9								
II 14-83	493	2.8								-102
II 14-83	519	3.2								-98
II 14-83	528	2.5		4.9	2.8	3.0				
II 14-83	551	0.7			1.1	3.4		-84		
II 14-83	559	0.3				3.6		-107		
II 14-83	569	0.1						-105		
II 14-83	573	0.3		5.1				-106		
II 14-83	589	1.0						-90		
II 14-83	598	2.2								
II 14-83	610	3.0				3.1		-115		
II 14-83	620	1.2								
II 14-83	634	2.9				2.2		-122		
II 14-83	644	1.2								
II 14-83	656	2.1								
II 14-83	668	2.5				4.3		-91		
II 14-83	681	1.0								
II 14-83	702	1.7								
II 14-83	722	2.0				3.3		-123		
II 14-83	751	1.9				2.7		-89		
II 14-83	770	1.1								
II 14-83	783	1.9								
II 14-83	790	2.2								
II 14-83	802	2.0		5.9				-75		
II 14-83	809	1.8								
II 14-83	817	-0.5								
II 14-83	822	0.9								
II 14-83	840	0.5								-87
II 14-83	854	1.5		5.8				-87		
II 14-83	867	8.3								
II 48-136	93		-11.7			2.7		-88		
II 48-136	107	2.0								
II 48-136	114	-0.1								-73
II 48-136	124					2.0		-87		
II 48-136	130	1.6								-120

TABLE 2. (Cont.)

Drill core	Depth (m)	VCDT (‰)		VSMOW (‰)						
		$\delta^{34}\text{S}$ (po-cp)	$\delta^{34}\text{S}$ (py)	$\delta^{18}\text{O}$ (ol)	$\delta^{18}\text{O}$ (pl)	$\delta^{18}\text{O}$ (se)	$\delta^{18}\text{O}$ (am)	δD (se)	δD (am)	δD (wr)
II 48-136	158	-0.3								
II 48-136	175					2.7				
II 48-136	188	-1.3								
II 48-136	231	0.3								
II 48-136	238	2.1								
II 48-136	241	3.8								
II 48-136	255	4.0								
II 48-136	294			4.3						
II 48-136	320		-8.9							
II 48-136	323	1.5								
II 48-136	340	0.7								
II 48-136	351	-0.4								
II 48-136	395	-4.0 (+py)								
II 48-136	410	1.2		3.5		3.3		-108		
II 48-136	442		-7.2					-93		
II 48-136	450	1.6								
II 48-136	461	1.5		3.4						
II 48-136	465	1.8								
II 48-136	473	2.6		4.1						
II 48-136	476	-0.1								-108
II 48-136	486			4.2						
II 48-136	497									-117
II 48-136	505	3.0		3.5						
II 48-136	516	2.7								
II 48-136	528	2.4								
II 48-136	538	1.1		4.1						
II 48-136	548	1.9								
II 48-136	552	-3.5 (+py)								
II 48-136	557	1.1								
II 48-136	568	1.9		4.4		3.1		-88		
II 48-136	577	1.1		5.2						
II 48-136	588	-4.3 (+py)				2.7				
II 48-136	599	0.8		3.5						
II 48-136	607	1.7								
II 48-136	622	1.5		3.9						
II 48-136	623	0.7								
II 48-136	630	1.2		4.8						
II 48-136	634	2.9				2.1				
II 48-136	646	0.5								
II 48-136	651	0.0								
II 48-136	655	1.3								
II 48-136	659	0.3		5.4						

Abbreviations: am = amphibole, cp = chalcopyrite, ol = olivine, pl = plagioclase, po = pyrrhotite, py = pyrite, ser = serpentine, wr = whole rock

balanced by the rate of new mineral growth (e.g., Merino and Dewars, 1998). Solid-state processes are in no way implied. An isovolumetric reaction of the form



may be written, which highlights the requirements for Mg and Si transfer. Local hydrolysis of olivine, pyroxene, and plagioclase provides the Mg^{2+} and Si^{4+} .

Isovolometric replacement reactions similar to equation (1) also can be written in terms of H_2S or HS^- , reflecting relatively lower f_{O_2} that is often associated with serpentinites (e.g., Frost, 1985). However, Frost (1985) has shown that for talc-carbonate-serpentine assemblages, f_{O_2} is commonly

within the sulfate stability field. Sulfate is considered important here because of the observed isotopic values of late-stage pyrite (see below) and because sulfide dissolution is enhanced where sulfate is the predominant aqueous sulfur species.

Isovolometric replacement of sulfides is indicated by the sharp contacts at grain boundaries (nonpenetrative fractures) and the lack of deformation in adjacent silicate grains (primarily partially serpentinitized olivine). Although the observed textures could be interpreted as the filling of fractures in sulfide minerals by serpentine, chlorite, and magnetite, the opening of fractures in the interstitial sulfide grains, whether by the force of crystallization exerted by the silicates or some other mechanism, must be accommodated by strain that would have affected adjacent mineral grains. This is not observed at the thin section scale (Figs. 4–6). Late-stage veins of pyrite-carbonate-magnetite-serpentine-chlorite (Fig. 6B) are

penetrative and crosscut both sulfide and silicate mineral assemblages. These veins are distinctly different from the non-penetrative veins in sulfide minerals that we interpret as replacement features.

It is very unlikely that the veins of silicate minerals in the interstitial sulfides represent trapped silicate melt. The volatile content of a mafic silicate melt would be insufficient to promote complete serpentinization, and fluid infiltration would be required to produce the serpentine $\pm$ chlorite assemblage. Because immiscible sulfide remains liquid after crystallization of the coexisting mafic silicate melt (e.g., Naldrett, 1989), a silicate liquid could not account for the presence of veinlike olivine and pyroxene in sulfide. The interpretation that the Jinchuan intrusion was emplaced as a viscous crystal mush (DeWaal et al., 2004; Li et al., 2004) also is consistent with the coexistence of olivine grains and interstitial sulfide liquid.

Area measurements indicate that between 6 and 34 percent of interstitial sulfide assemblages have been replaced by silicate and oxide minerals, with an average of ~ 10 percent in all samples. Some of the sulfides that have been removed from the net-textured assemblages appear to have been locally re-deposited in fine ribbons and veinlets in nearby serpentine that replaces olivine. Sulfide stringers also are found locally in chlorite and amphibole. However, the irregular distribution of fine stringers and veinlets that represent remobilized sulfide make a quantitative assessment of mass transfer difficult. Some of the metals lost from the sulfide assemblage have been taken up by serpentine and chlorite (see above), but this accounts for less than 1 percent of the metals lost during replacement of the sulfides. The fine-grained sulfides present within serpentine, chlorite, and amphibole are also insufficient to balance those removed from the sulfide assemblages. Pentlandite is more extensively replaced than either pyrrhotite or chalcopyrite (see above), so Ni mobility may have been as great as that of Cu, even though chalcopyrite is five times more abundant than pentlandite.

Sulfur isotope values

For many large magmatic Ni-Cu sulfide deposits externally derived sulfur was important for promoting sulfide saturation in mafic melts at relatively low pressure (e.g., Grinenko, 1985; Mavrogenes and O'Neill, 1999; Ripley et al., 1999, 2002, 2003; Holzheid and Grove, 2002). It is, therefore, somewhat surprising that direct evidence for sulfur assimilation at Jinchuan is lacking. The bulk of the data fall in the range of 0 ± 2 per mil, which would be consistent with sulfur being of mantle origin. However, locally higher $\delta^{34}\text{S}$ values from net-textured sulfide assemblages suggest that at least some non-mantle-derived sulfur may have been involved in ore deposition. Immediate country rocks (granitoids, gneiss, and marble) have a low sulfur content (generally <100 ppm) and appear to be poor candidates as sulfur sources. We have found three gneiss samples with elevated sulfur as pyrite with $\delta^{34}\text{S}$ values ranging from 2.5 to 7.7 per mil. Sulfide-bearing shales may exist in the stratigraphic section at Jinchuan, but none are exposed.

The strongly negative $\delta^{34}\text{S}$ values of pyrite in the deposit may be interpreted in at least three ways. One process which is consistent with the textural evidence for sulfur loss from the sulfide assemblages as well as the production of secondary

magnetite is the generation of oxidized sulfur by a reaction with oxygen-rich fluid (e.g., reaction 1). Partial reduction of sulfate by reaction with ferrous iron or hydrogen from elsewhere in the deposit, or possibly due to reaction with sulfate-reducing bacteria, could lead to the generation of secondary pyrite with a large associated isotopic fractionation at temperatures less than 400°C (Fig. 8). Between ~ 10 and 50 percent reduction of SO_4^{2-} with a $\delta^{34}\text{S}$ value near 0 per mil can produce H_2S with $\delta^{34}\text{S}$ values between ~ -25 and -10 per mil (Ohmoto, 1972; Fig. 8). These values are very near those of pyrite found in veins within the deposit.

Aqueous sulfide with strongly negative isotopic compositions also could be produced at somewhat lower f_{O_2} conditions where both sulfate and sulfide are products of sulfide mineral dissolution. The production of both oxidized and reduced aqueous sulfur species would lead to strong isotopic fractionation under conditions close to sulfide-sulfate equilibrium (Ohmoto, 1972; Fig. 8). Isotopically negative H_2S could then be involved in the generation of secondary pyrite in the deposit.

An external source for the light sulfur in secondary pyrite seems unlikely, given that none is recognized in the exposed country rocks and especially considering the evidence for the

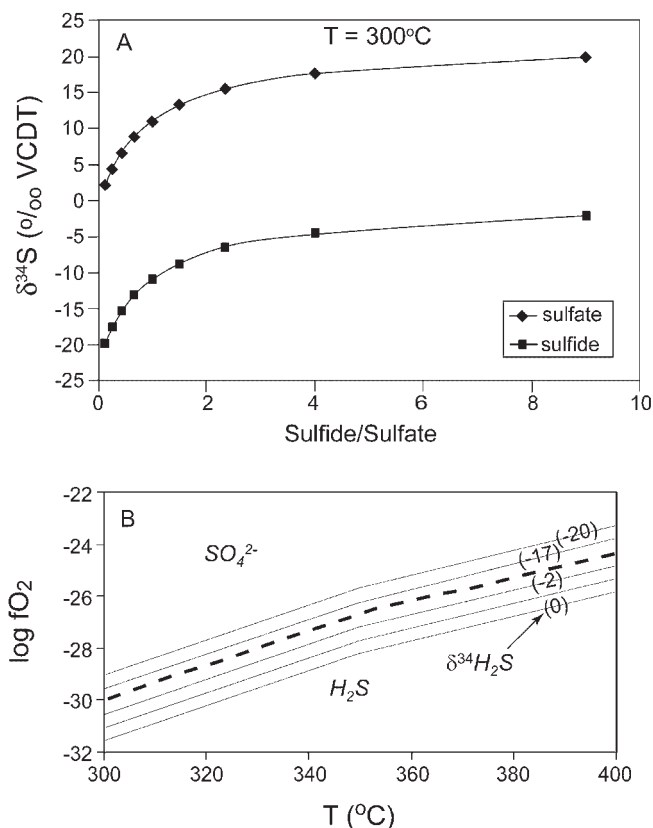


FIG. 8. A. Sulfur isotope values of sulfate and sulfide produced during the partial reduction of sulfate with a $\delta^{34}\text{S}$ value of 0 per mil. The Δ value for SO_4^{2-} - H_2S at 300°C is ~ 22 per mil. B. $\log f_{\text{O}_2}$ versus temperature diagram illustrating the variation in $\delta^{34}\text{S}_{\text{H}_2\text{S}}$ values near the sulfate and/or sulfide boundary. Strongly negative $\delta^{34}\text{S}$ values are produced at mole fraction of H_2S values less than 0.5. The pH for the calculations was taken as neutral; this has no effect on the relative $\delta^{34}\text{S}$ values (Ripley and Ohmoto, 1977; Ohmoto, 1986).

loss of sulfur from the net-textured ores of the Jinchuan deposit during hydrothermal alteration.

Oxygen and hydrogen isotope evidence for fluid sources

Oxygen and hydrogen isotope compositions of minerals from the Jinchuan intrusion confirm a history involving extensive interaction with fluid. Olivine $\delta^{18}\text{O}$ values indicate that mantle signatures of 5 to 6 per mil are preserved in samples where <40 percent of the olivine has been serpentinized. As the intensity of serpentinization increases, $\delta^{18}\text{O}$ values of olivine decrease. Although only four samples of plagioclase could be measured because of alteration and low modal abundance, none of the samples have values of 5.5 to 6 per mil that are typical of plagioclase from pristine mantle-derived basaltic magmas. The extremely low $\delta^{18}\text{O}$ values of 3.1 to 1.1 per mil indicate extensive oxygen isotope exchange with a fluid depleted in ^{18}O . Although only two samples with coexisting plagioclase and olivine could be measured, the disequilibrium ^{18}O fractionations indicate kinetically limited isotopic exchange.

Because our sampling has been restricted to only three drill cores we lack the spatial control that is necessary to rigorously evaluate isotopic fractionation and mass transport. However, the formulation of Gregory and Criss (1986) for fluid-buffered systems allows an estimation of effective fluid/rock ratios using coexisting plagioclase and olivine. Although only two sample pairs could be analyzed, the steep slope on a $\delta^{18}\text{O}_{\text{plagioclase}}$ versus $\delta^{18}\text{O}_{\text{olivine}}$ plot suggests that plagioclase has undergone extensive oxygen isotope exchange with a relatively low ^{18}O fluid, whereas olivine has not. We estimate fractional approach to equilibrium, or f values, for plagioclase of ~ 0.8 and for olivine of ~ 0.2 . For a rock with 90 percent olivine and 10 percent plagioclase these values render a time-integrated fluid/rock ratio of ~ 4 on a weight basis (Gregory and Criss, 1986), assuming that the ratio of isotopic exchange rates for plagioclase and olivine is ~ 100 . What is important here is not the absolute values, but the fact that the estimated fluid/rock ratio is considerably higher than that required for the stoichiometric conversion of olivine to serpentine ($w/r \sim 0.14$). Taking the lowest plagioclase $\delta^{18}\text{O}$ value of 1.1 as an equilibrium value, the $\delta^{18}\text{O}$ value of water with which plagioclase ($\sim \text{An } 30\text{--}40$) equilibrated at 350°C is ~ 0 per mil using the fractionation factor of Zheng (1993b). Water $\delta^{18}\text{O}$ values calculated from amphibole and serpentine $\delta^{18}\text{O}$ values at 300° to 400°C range from 3.0 to 5.8 and 1.7 to 4.3 per mil, respectively, using the fractionation factors of Zheng (1993a).

The δD values of H_2O in equilibrium with amphibole at temperatures between 300° and 400°C range from -52 to -63 per mil, using the hornblende-water fractionation factor of Graham et al. (1984). The δD of water in equilibrium with serpentine between 300° and 400°C has a wide range of δD values from -45 to -103 per mil, using the fractionation factors of Wenner and Taylor (1973). The small range of $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ values computed from amphibole and serpentine but the large range in computed $\delta\text{D}_{\text{H}_2\text{O}}$ values may be interpreted as a result of isotopic exchange involving low integrated water/rock ratios or, alternatively, that at least two distinct fluids were involved in the alteration process at Jinchuan. Although absolute values of computed water δD values depend on the choice of mineral- H_2O fractionation

factors, our interpretation of multiple fluids involved in the alteration process (see below) holds regardless of the particular mineral- H_2O fractionation factor utilized.

The fluid paths illustrated in Figure 9 represent isotopic exchange with rocks at variable "effective" or "time-integrated" water/rock ratios. The low water/rock ratios indicated along the fluid paths shown in Figure 9 can also represent isotopic exchange involving fluids that have exchanged with rocks over a long path length. Unevolved, short path length fluids are more characteristic of "fluid-buffered" systems and high time-integrated water/rock ratios. The point here is that the relatively low water/rock ratios suggested by the isotopic compositions of waters calculated to be in equilibrium with serpentine (Fig. 9) may apply to fluids that had undergone extensive isotopic exchange in a flow system prior to interaction with olivine-rich rocks at Jinchuan.

Figure 9 shows that the water in equilibrium with amphibole may have been meteoric water that had exchanged with basaltic crust ($\text{H}_2\text{O} = 1$ wt %, $\delta^{18}\text{O} \sim 6\text{‰}$, $\delta\text{D} \sim -60\text{‰}$) at low water/rock ratios or a mixture of an evolved meteoric water (low δD) with either a metamorphic fluid or possibly evolved seawater (high δD). The similarity in the computed isotopic values of H_2O in equilibrium with amphibole to the high δD population in equilibrium with the serpentine suggests similar sources of water.

The lowest δD values of water in equilibrium with serpentine at Jinchuan strongly suggest that meteoric water was involved in either primary serpentine deposition or later exchange reactions. Kyser et al. (1999) have also suggested that the large range in δD values, along with relatively constant $\delta^{18}\text{O}$ values, recorded by many serpentinites is a result of continued late-stage hydrogen isotope exchange with meteoric waters. Continued hydrogen isotope exchange with meteoric water having a low δD is a convenient mechanism to explain the range in δD values found in our data set. The presence of

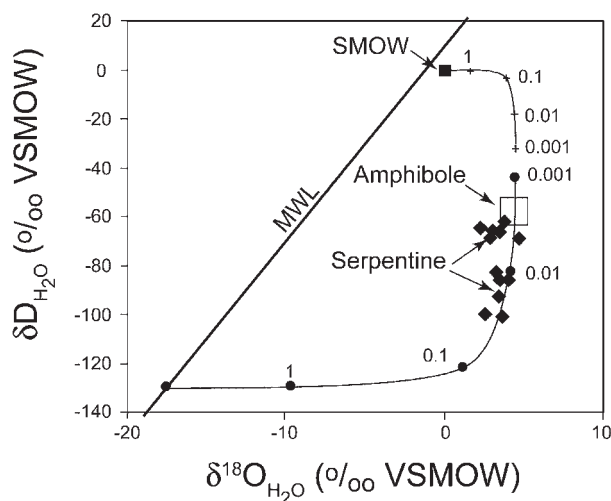


FIG. 9. $\delta^{18}\text{O}_{\text{H}_2\text{O}}$ vs. $\delta\text{D}_{\text{H}_2\text{O}}$ plot showing the isotopic compositions of water in equilibrium with amphibole (box) and serpentine from Jinchuan at 350°C . The illustrated exchange paths for seawater and a meteoric water at various water/rock weight ratios are computed for exchange with a basaltic rock with $\delta^{18}\text{O} = 6$ per mil, $\delta\text{D} = -60$ per mil, 1 wt percent H_2O , and $\Delta^{18}\text{O}$ (rock- H_2O) and ΔD (rock- H_2O) of 1.5 and -25 percent, respectively, at 350°C . MWL is the global meteoric water line (Ohmoto, 1986).

late-stage serpentine-chlorite-magnetite veins that crosscut serpentized assemblages is evidence for possible late-stage hydrogen isotope exchange.

Summary and Conclusions

Both mineralogical and stable isotope data suggest a complex history of hydrothermal alteration of the Jinchuan intrusion and its contained sulfide ores. The replacement textures in sulfide minerals at Jinchuan are unusual when compared to those in other deposits such as Duluth (Ripley and Alawi, 1986), Voisey's Bay (Naldrett et al., 1996), and Noril'sk (Li et al., 2003). The primary difference is the degree of interaction with hydrothermal fluids in the Jinchuan intrusion. Olivine is found in various states of replacement by serpentine, magnetite, chlorite, and amphibole, and interstitial pyroxene and plagioclase have been locally converted to chlorite, amphibole, epidote, and more sodic plagioclase. Interstitial and net-textured sulfide minerals also have been replaced by chlorite, serpentine, and magnetite. Locally as much as 30 percent of the magmatic sulfide assemblage has been replaced by silicate and oxide minerals, with an average of 10 percent. Pyrite in late-stage calcite-pyrite-serpentine-magnetite veins is characterized by negative $\delta^{34}\text{S}$ values, consistent with a partial reduction of sulfate produced during fluid-induced oxidation of the primary sulfides and reprecipitation in the late pyrite veins.

Oxygen and hydrogen isotope studies of silicate minerals indicate that observed isotopic compositions are the result of rock interaction with at least two distinct fluids (high and low δD). Oxygen and hydrogen isotope compositions of amphibole and oxygen isotope compositions of feldspar and serpentine are all consistent with the involvement of evolved meteoric water or seawater. Although a metamorphic fluid cannot be ruled, the low H_2O content of preintrusion H_2O -poor metamorphic rocks in the area and the high water/rock ratio required for the alteration and indicated by isotopic relationships are more consistent with the involvement of a large fluid reservoir. The range in serpentine δD values may be explained in at least two ways. Serpentine may have formed in equilibrium with a fluid that was derived by varying degrees of mixing of an evolved meteoric water and either a metamorphic water or seawater. Alternatively, late-stage hydrogen isotope exchange with meteoric water could account for the low δD of water in equilibrium with some serpentine.

The large volume of fluid that appears to have exchanged with ultramafic rocks at Jinchuan is consistent with large-scale hydrothermal convection and a rift-related origin for the magmatism. Chai and Naldrett (1992b) proposed a continental rift setting for the Jinchuan intrusion and speculated that rift-related tensional faults and mafic volcanic rocks are concealed, or in the case of the volcanic rocks, eroded away. In such environments, the extensive subsolidus alteration history of the deposit must be considered when using metal ratios or metal-to-sulfur ratios to evaluate the primary mineralization.

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